

1 Statistical Issues in Multicenter Randomized Clinical
2 Trials

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5 **Abstract**

6 Multicenter randomized controlled trials require a choice among ignor-
7 ing, fixing, or randomizing site effects in the primary analysis, and this
8 choice interacts with stratified-randomization adjustment, treatment-by-site
9 interaction, and site-size balance. We report a factorial Monte Carlo sim-
10 ulation, structured under the ADEMP framework of Morris, White, and
11 Crowther (2019), that jointly varies the number of sites, site-level variance,
12 treatment-by-site interaction, and site-size balance, comparing OLS ignoring
13 site, OLS with fixed site effects, and a linear mixed model with random site
14 effects. In addition to bias, empirical and model-based standard error, power,
15 and coverage under a true alternative treatment effect, we report empirical
16 Type I error under a true null effect, and singular/boundary-fit rates for
17 the random-effects model, both previously unreported in this simulation de-
18 sign. The random-effects model attains the best combination of power and
19 coverage across most conditions; ignoring site produces conservative (not
20 anti-conservative) inference under stratified randomization, consistent with
21 prior work; and all three methods control Type I error near the nominal level
22 in the null-hypothesis scenarios simulated. Every point estimate is reported
23 with its Morris et al. Table 6 Monte Carlo standard error. Code and a full
24 ADEMP audit trail accompany the simulation.

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52 1 Introduction

53 Multicenter randomized clinical trials (RCTs) represent the standard design for
54 confirmatory evaluation of therapeutic interventions. By enrolling participants
55 across multiple clinical sites, these trials achieve practical recruitment targets
56 while broadening the population base for generalization of findings^{1,2}. However,
57 the multicenter structure introduces several analytic complexities that, if mishan-
58 dled, can distort inference about treatment effects. We address four interrelated
59 statistical issues that arise in the design and analysis of multicenter RCTs with
60 continuous outcomes, and we present simulation evidence bearing on each in turn.

61 1.1 Modeling site effects

62 The first question concerns how site (or center) effects should be incorporated
63 into the primary analysis. Three broad strategies exist: ignoring site differences
64 entirely, treating sites as fixed effects, and treating sites as random effects drawn
65 from a population of potential sites. Each strategy carries distinct assumptions
66 and implications for the validity and efficiency of treatment effect estimation^{3,4}.

67 Localio and colleagues provided an influential overview of the problem, identifying
68 three analytic challenges inherent to multicenter data: within-center correlation
69 of outcomes, confounding by center, and effect modification across centers². The
70 choice between fixed and random effects models has been debated extensively.
71 Fixed effects models condition on the observed sites and make no distributional
72 assumptions about site effects, but they consume degrees of freedom rapidly when

the number of sites is large relative to total sample size⁵. Random effects models treat site effects as realizations from a normal distribution, estimating only a variance component rather than individual site parameters, and thereby preserve statistical efficiency⁶. The ICH E9 guideline recommends that mixed models are ‘especially relevant when the number of sites is large’¹.

Chu and colleagues conducted a comprehensive simulation comparing six analytic approaches for continuous outcomes across varying numbers of centers, center sizes, and intraclass correlation coefficients (ICCs)⁷. All methods yielded unbiased point estimates, but ignoring centers inflated the standard error and reduced power when within-center correlation was present. The mixed-effects model attained nominal coverage and near-optimal power across nearly all configurations. Generalized estimating equations (GEE) underestimated the standard error when the number of centers was small.

Kahan and Morris extended these findings, demonstrating that random center effects performed at least as well as fixed effects in all scenarios examined, and substantially outperformed them when the number of patients per center was small⁸. Islam and Bangdiwala confirmed these patterns in a recent replication study with both continuous and binary outcomes, reinforcing the recommendation for random effects models as a default analytic strategy⁹.

1.2 What does “random” actually mean here?

Before going further, it is worth being precise about the word “random,” because this word carries most of the weight in the discussion above, and because it can mean two different things. We spell out both meanings in plain terms, with the underlying math shown alongside, since the two meanings lead to two different arguments later in this Introduction.

1.2.1 A hospital-specific shift, fixed or random

Picture a trial run at ten hospitals. Each hospital has its own typical outcome level, for reasons that have nothing to do with the treatment being tested: a different mix of patients, different routine practices, different equipment. Call that hospital-specific shift α_i , for hospital i . A simple model for one patient’s outcome, written Y_{ij} for patient j at hospital i , looks like this:

$$Y_{ij} = \alpha_i + \delta T_{ij} + \epsilon_{ij}$$

Here T_{ij} is 1 if that patient received the active treatment and 0 if not, δ is the treatment effect the trial wants to estimate, and ϵ_{ij} is ordinary patient-to-patient variation that has nothing to do with which hospital a patient attended.

The open question is what kind of quantity α_i is.

There are two options. The first treats each hospital’s shift as an ordinary, fixed number, no different in kind from a patient’s age or sex: something to adjust for,

110 one number per hospital. Ten hospitals means ten fixed numbers to estimate.

111 The second option treats α_i as random, drawn from a normal distribution: $\alpha_i \sim$
112 $N(0, \sigma_\alpha^2)$. Under this option, the one unknown quantity of real interest is σ_α^2 : how
113 much do hospitals typically differ from one another? Rather than estimating ten
114 separate numbers, the analysis estimates a single number describing the *spread* of
115 hospital effects.

116 This is not a small technical difference. It changes what the analysis is entitled
117 to claim. Estimating ten fixed numbers tells you only about these ten specific
118 hospitals. Estimating one spread, σ_α^2 , is a broader claim: it says something about
119 hospital-to-hospital variation in general, of the kind you would expect if the trial
120 had, by chance, enrolled a different set of ten hospitals instead.

121 That broader claim is exactly where a disagreement in the published literature
122 shows up, and we come back to it below.

123 **1.2.2 Why the simulation needs a genuinely random site effect**

124 This paper is a computer simulation. It generates many pretend trials, over and
125 over, and checks how well each analysis strategy (ignore site, treat site as fixed,
126 treat site as random) recovers the true treatment effect and reports honest uncer-
127 tainty around it. For that check to mean anything, the computer program that
128 generates each pretend trial has to actually build in hospital-to-hospital variation,
129 and it has to do so afresh for every pretend trial, not use the same ten fixed
130 numbers every time.

131 That is what drawing $\alpha_i \sim N(0, \sigma_\alpha^2)$ inside the simulation accomplishes: every
132 pretend trial gets its own new set of hospital shifts, all coming from a distribution
133 with the same spread, σ_α^2 . Only by rebuilding this variation from scratch each
134 time can the simulation ask a fair question: across many possible versions of “a
135 multicenter trial like this one,” does each analysis strategy give honest confidence
136 intervals and adequate power? A simulation that reused the same fixed hospital
137 numbers every time would not be testing that question at all; it would only be
138 testing how well each method recovers one specific, already-fixed set of numbers.
139 The later sections on treatment-by-site interaction and on imbalanced site sizes
140 work the same way: each also needs a freshly drawn random ingredient, every
141 pretend trial, for the same reason.

142 It is important to be clear about what this operational choice does, and does
143 not, claim. It is a requirement of building a fair test, not a statement about how
144 real hospitals were chosen for a real trial. That second, separate question is the
145 subject of the next section.

146 **1.2.3 Two different reasons statisticians give for treating site as ran-** 147 **dom**

148 Everything so far concerns how the simulation is built. A separate question is why,
149 in a real trial with real hospitals, an analyst should choose to treat site as random

rather than fixed when fitting the actual statistical model. Published statisticians do not fully agree on the reason, even where they agree on the recommendation, and a statistically literate reader deserves to know both arguments.

The first argument says the random-effects model literally means what it appears to mean: it treats the hospitals in the trial as if they had been drawn at random from a much larger population of possible hospitals. Falissard and Chavance make this point directly: the random-effects interpretation is only truly justified when the hospitals really were chosen at random from a wider population¹⁰. In practice, that is almost never how hospitals are chosen. Hospitals join a trial because they can recruit patients quickly, because an investigator has a relationship with the study team, or because of funding and logistics, not by a lottery over all possible hospitals³.

The second argument does not depend on how the hospitals were chosen at all. Kahan and Morris justify adjusting for site on the design of the trial itself: patients were randomly split between treatment and control separately within each hospital, so the analysis should reflect that split, no matter how the hospitals themselves came to be in the trial^{8,11}. Edgar, Roberts, and Sharples reach the same conclusion by re-analyzing a real trial run across 271 hospitals in 40 countries: they recommend a random site effect for practical reasons, better power and less bias, again without any claim about hospitals being a random sample of a larger population¹².

These two arguments do not agree on *why* a random site effect is the right choice, but they agree on *what to do*: prefer a random effect for site once there are more than a handful of hospitals. The recommendation in this paper rests on that agreement, so it stands regardless of which argument a reader finds more convincing. The simulation itself follows the second, design-based tradition in spirit: it builds hospital-to-hospital variation into every pretend trial as an operational necessity for testing the analysis strategies fairly (see above), without needing or claiming that real hospitals are ever a literal random sample of all possible hospitals.

1.3 Failure to adjust for stratification variables

The second issue concerns the consequences of ignoring stratification variables, including site, in the analysis. When randomization is stratified by center (as is standard in multicenter trials), failing to adjust for the stratification factor in the analysis leads to biased inference. Kahan and Morris demonstrated through simulation that an unadjusted analysis of a trial randomized with stratified blocks produces standard errors that are biased upward, confidence intervals that are too wide, type I error rates below the nominal level, and a loss of statistical power¹¹. This paradoxical conservatism arises because the stratified randomization induces negative correlation between treatment groups within strata; ignoring this correlation inflates variance estimates.

Kahan and Morris extended this work to trials with multiple stratification factors, finding that adjusting for the main effects of all stratification variables was gener-

ally sufficient without requiring adjustment for all cross-classified strata¹³. These findings align with the ICH E9 recommendation that ‘the statistical model used for analysis should reflect the restriction imposed by the randomisation procedure’¹, and with current regulatory guidance on covariate adjustment more broadly¹⁴.

1.4 Treatment-by-site interaction

The third concern is heterogeneity of the treatment effect across sites. Even under a common protocol, eligible participants, site practices, and investigator experience may vary, producing genuine treatment-by-site interaction¹⁵. Senn argued that some degree of treatment-by-center interaction is inevitable in practice and that a rational approach to planning multicenter trials should anticipate it³.

The ICH E9 guideline recommends first fitting a model without the interaction term to estimate the main treatment effect, and then examining heterogeneity as a secondary analysis. However, the power to detect treatment-by-site interaction is typically low, so a non-significant interaction test does not establish homogeneity¹⁶. Jones and colleagues compared fixed-effects weighted estimators, unweighted estimators, and random effects estimators in the presence of varying degrees of treatment-by-center interaction⁴. The fixed effects weighted estimator performed well across a range of interaction magnitudes, while unweighted estimators could be inefficient when site sizes differed markedly.

When treatment-by-site interaction is modeled as random (i.e., the treatment slope varies across sites), the mixed model estimates both the average treatment effect and the variance of treatment effects across sites, providing a natural summary of heterogeneity¹⁷.

1.5 Imbalanced site sizes

The fourth issue, closely related to the preceding three, is whether imbalance in site sizes affects inference. Multicenter trials frequently exhibit highly unequal enrollment across sites, with a few large sites contributing a disproportionate share of participants. Senn noted that a rational approach to planning multicenter trials leads naturally to unequal site sizes, because sites with higher recruitment capacity will enroll more patients within the trial’s fixed time window³.

Ruvuna introduced a coefficient of imbalance to quantify the power loss attributable to unequal center sizes and showed that substantial imbalance can reduce power when an unweighted (Type III) analysis is used¹⁸. Vierron and Giraudeau formalized this through a design effect framework, demonstrating that the efficiency loss depends on both the ICC and a statistic quantifying the heterogeneity of group distributions across centers¹⁹. When the ICC is positive, balanced allocation is optimal, but moderate imbalance imposes only a modest efficiency penalty. The key risk emerges when site size is informative (that is, when larger sites systematically differ from smaller sites in patient characteristics or treatment delivery), potentially introducing bias into the treatment effect estimate^{15,20}.

1.6 Present study

The existing literature provides clear theoretical guidance but limited integrated simulation evidence spanning all four issues simultaneously. We present a Monte Carlo simulation that jointly varies the analytic method (ignore site, fixed site effects, random site effects), the presence or absence of treatment-by-site interaction, balanced versus imbalanced site sizes, and the magnitude of site-level variance. The simulation evaluates bias, empirical standard error, model-based standard error, power, and coverage probability of 95% confidence intervals for the average treatment effect across all factorial combinations of these design features.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther²¹, the simulation is reported under the ADEMP framework.

- **Aims.** Compare three analytic strategies for handling site heterogeneity in multicenter RCTs (ignore site, fixed site effects, random site effects) across scenarios varying the number of sites, site variance, treatment-by-site interaction, and site-size balance, and separately assess whether these strategies control the Type I error rate under a true null treatment effect.
- **Data-generating mechanism.** Detailed in the next subsection.
- **Estimand.** The overall treatment effect δ on the outcome scale, in the simulation-performance sense of Morris et al. (a data-generating parameter with a known true value), not in the ICH E9(R1) sense of a fully specified population/variable/ intercurrent-event/summary-measure estimand²².
- **Methods.** OLS ignoring site, OLS with fixed site effects, and a linear mixed model with random site effects via `lme4::lmer`. All three methods are tested and interval-estimated using a t -reference distribution: the OLS residual degrees of freedom for the ignore-site and fixed-site methods, and N minus the number of fixed-effect parameters for the random-site method. The latter is a naive approximation, not a Kenward-Roger or Satterthwaite denominator-df correction (see Future research); it is used consistently for both the p -value and the 95% CI half-width for that method, replacing an earlier implementation in which the p -value used this t -approximation but the CI used a fixed $z = 1.96$ half-width regardless of sample size or method.
- **Performance measures.** Bias, empirical SE, MSE, mean model SE, power at $\alpha = 0.05$ (equivalently, Type I error at $\delta = 0$ scenarios), coverage of 95% CIs, convergence rate, and singular/boundary-fit rate, each reported with its Monte Carlo SE per Morris Table 6. Non-convergence (a hard error from `lmer`) and singular/boundary fits (a successful fit at the edge of the random-effects covariance parameter space, which `lme4` flags as a warning rather than an error) are tracked as two distinct first-class outcomes and reported per scenario and method, rather than treating a fit that raised any warning as simply non-convergent or, conversely, absorbing a boundary fit

275 silently into “converged” with no record that it was a boundary solution.

276 2.2 Data generating model

277 For each simulated trial, data are generated from a two-level model with patients
278 nested within sites. Let Y_{ij} denote the continuous outcome for patient j at site i :

$$Y_{ij} = \alpha_i + (\delta + \gamma_i)T_{ij} + \epsilon_{ij}$$

279 where $\alpha_i \sim N(0, \sigma_\alpha^2)$ is the random site intercept, $\delta = 0.30$ is the true average
280 treatment effect, $\gamma_i \sim N(0, \sigma_\gamma^2)$ is the random treatment-by-site interaction, $T_{ij} \in$
281 $\{0, 1\}$ is the treatment indicator (balanced within each site via permuted blocks),
282 and $\epsilon_{ij} \sim N(0, \sigma_e^2 = 1)$ is the residual error.

283 2.3 Parameter values

284 The simulation crosses the following factors:

- 285 • **Number of sites:** $K = 10$ (few) vs $K = 30$ (many)
- 286 • **Site-level variance:** $\sigma_\alpha^2 \in \{0, 0.10, 0.50\}$ (none, moderate, large)
- 287 • **Treatment-by-site interaction:** $\sigma_\gamma^2 \in \{0, 0.04\}$ (none vs present)
- 288 • **Site size balance:** balanced ($n_i = 20$ for all i) vs imbalanced (site sizes
289 drawn from a Gamma(0.8, 0.8/20) distribution, yielding a coefficient of vari-
290 ation ≈ 1.1)

291 The true treatment effect is $\delta = 0.30$ across all scenarios, representing a moderate
292 standardized effect size of 0.30 standard deviations.

293 2.4 Analytic methods

294 Each simulated dataset is analyzed by three methods:

- 295 1. **Ignore site:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + e_{ij}$ (ordinary least squares, no site term)
- 296 2. **Fixed site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + \sum_{k=2}^K \beta_k I(i = k) + e_{ij}$ (site indicators
297 as fixed effects)
- 298 3. **Random site effects:** $Y_{ij} = \beta_0 + \beta_1 T_{ij} + u_i + e_{ij}$ where $u_i \sim N(0, \sigma_u^2)$
299 (linear mixed model via REML, fit with `lme4`)

300 2.5 Simulation procedure

301 Scenario 1/24: K10_noVar_bal ... Scenario 2/24: K10_modVar_bal ... Scenario
302 3/24: K10_hiVar_bal ... Scenario 4/24: K10_modVar_imbal ... Scenario
303 5/24: K10_hiVar_imbal ... Scenario 6/24: K10_modVar_int ... Scenario
304 7/24: K10_hiVar_int ... Scenario 8/24: K10_hiVar_int_imb ... Scenario
305 9/24: K30_noVar_bal ... Scenario 10/24: K30_modVar_bal ... Scenario 11/24:
306 K30_hiVar_bal ... Scenario 12/24: K30_modVar_imbal ... Scenario 13/24:
307 K30_hiVar_imbal ... Scenario 14/24: K30_modVar_int ... Scenario 15/24:
308 K30_hiVar_int ... Scenario 16/24: K30_hiVar_int_imb ... Scenario 17/24:

309 H0_K10_noVar_bal ... Scenario 18/24: H0_K10_modVar_bal ... Scenario
 310 19/24: H0_K10_hiVar_bal ... Scenario 20/24: H0_K10_hiVar_imbal ... Sce-
 311 nario 21/24: H0_K30_noVar_bal ... Scenario 22/24: H0_K30_modVar_bal ...
 312 Scenario 23/24: H0_K30_hiVar_bal ... Scenario 24/24: H0_K30_hiVar_imbal
 313 ...

314 Each scenario is replicated $R = 1,500$ times. The Monte Carlo SE of an estimated
 315 coverage or Type I error probability p is $MCSE = \sqrt{p(1-p)/R}$ (Morris, White,
 316 and Crowther 2019, Table 6); evaluated at the nominal $p = 0.95$ (worst case
 317 among the coverage targets considered here), $R \geq 1,320$ is required for $MCSE$
 318 ≤ 0.6 percentage points, and $R = 1,500$ gives $MCSE \approx 0.56$ pp, satisfying the
 319 target with a margin. Treatment is assigned via balanced allocation within each
 320 site (equal numbers of treatment and control participants, or within ± 1 when n_i
 321 is odd). All random number generation uses a fixed seed (`set.seed(20260310)`)
 322 and the L'Ecuyer-CMRG RNG for reproducibility. The `sim-run` chunk's cache is
 323 keyed on the MD5 hash of `analysis/scripts/sim_multicenter.R` in addition to
 324 the chunk's own text, so a change to the simulation functions themselves (not just
 325 to this chunk) correctly invalidates the cached result rather than silently reusing
 326 output from a prior version of the simulation code.

327 3 Results

328 3.1 Headline findings

329 We observe that across 16 alternative- hypothesis scenarios ($\delta = 0.30$) and $R =$
 330 1,500 replications per scenario, the random-site mixed model attained power be-
 331 tween 0.54 and 0.97, with coverage of the 95% confidence interval ranging from
 332 0.895 to 0.960 and absolute bias never exceeding 0.011 (Morris Table 6 Monte
 333 Carlo SEs are given in parentheses next to the power and coverage point estimates
 334 in Table 1). The fixed-site estimator closely tracked the random-site estimator.
 335 Ignoring site entirely produced overcoverage (up to 0.984) and correspondingly
 336 lower power (as low as 0.40) in scenarios with non-trivial between-site variance.
 337 The lowest convergence rate (hard `lmer` errors only) observed across any scenario
 338 or method was 1.000; however, the random-site model's *singular/boundary* fit rate
 339 reached as high as 0.557 in the low- and no-site-variance scenarios, where the true
 340 random-intercept variance is at or near the parameter-space boundary. A singular
 341 fit still returns a usable point estimate, but its variance-component estimate is
 342 degenerate; see "Type I error under the null" below and the Discussion for the
 343 implications. Under the null-hypothesis scenarios ($\delta = 0$; Table 2), the empirical
 344 Type I error rate for the random-site method ranged from 0.041 to 0.068, and for
 345 the ignore-site method from 0.014 to 0.047, against a nominal $\alpha = 0.05$.

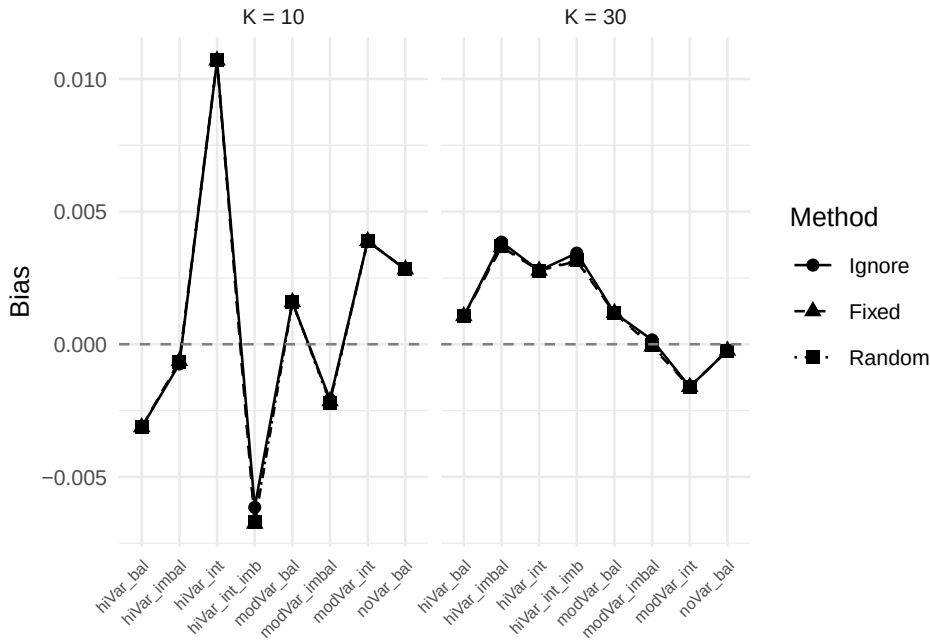


Figure 1: Bias of the treatment effect estimate across scenarios and analytic methods. Dashed line at zero indicates no bias.

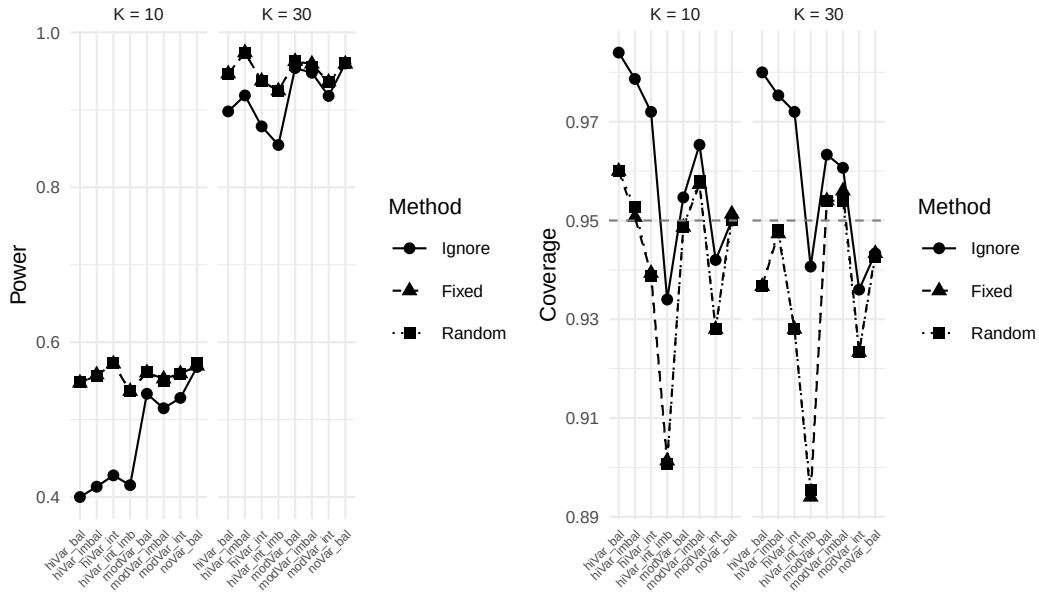


Figure 2: Power (left) and coverage probability (right) across scenarios. Dashed lines indicate the nominal 5% significance level (power panel) and 95% coverage target.

Table 1: Simulation results under the alternative hypothesis ($\delta = 0.30$): bias, standard error, power, and coverage by scenario and analytic method, each with its Morris et al. (2019) Table 6 Monte Carlo SE in parentheses where applicable. R = 1500 replications per scenario. ‘Conv.’ is the rate of successful `lmer` fits (errors only); ‘Sing.’ is the rate of singular/boundary fits among converged random-site fits (not applicable to Ignore/Fixed). Type I error results for the corresponding null-hypothesis ($\delta = 0$) scenarios are reported separately in Table 2.

Scenario	Method	Bias	Emp. SE	Model SE	Power (MCSE)	Coverage (MCSE)	Conv.	Sing.
K10 hiVar bal	Fixed	-0.003	0.140	0.141	0.547 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar bal	Ignore	-0.003	0.140	0.169	0.400 (0.013)	0.984 (0.003)	100.0%	0.0%
K10 hiVar bal	Random	-0.003	0.140	0.141	0.548 (0.013)	0.960 (0.005)	100.0%	0.0%
K10 hiVar imbal	Fixed	-0.001	0.141	0.141	0.558 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 hiVar imbal	Ignore	-0.001	0.142	0.167	0.413 (0.013)	0.979 (0.004)	100.0%	0.0%
K10 hiVar imbal	Random	-0.001	0.141	0.141	0.557 (0.013)	0.953 (0.005)	100.0%	0.1%
K10 hiVar int	Fixed	0.011	0.148	0.142	0.572 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int	Ignore	0.011	0.148	0.171	0.428 (0.013)	0.972 (0.004)	100.0%	0.0%
K10 hiVar int	Random	0.011	0.148	0.142	0.573 (0.013)	0.939 (0.006)	100.0%	0.0%
K10 hiVar int imb	Fixed	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.0%
K10 hiVar int imb	Ignore	-0.006	0.173	0.168	0.415 (0.013)	0.934 (0.006)	100.0%	0.0%
K10 hiVar int imb	Random	-0.007	0.173	0.142	0.537 (0.013)	0.901 (0.008)	100.0%	0.2%
K10 modVar bal	Fixed	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	0.0%
K10 modVar bal	Ignore	0.002	0.141	0.148	0.533 (0.013)	0.955 (0.005)	100.0%	0.0%
K10 modVar bal	Random	0.002	0.141	0.141	0.561 (0.013)	0.949 (0.006)	100.0%	3.2%
K10 modVar imbal	Fixed	-0.002	0.139	0.141	0.553 (0.013)	0.957 (0.005)	100.0%	0.0%
K10 modVar imbal	Ignore	-0.002	0.139	0.147	0.515 (0.013)	0.965 (0.005)	100.0%	0.0%
K10 modVar imbal	Random	-0.002	0.139	0.141	0.550 (0.013)	0.958 (0.005)	100.0%	8.2%
K10 modVar int	Fixed	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	0.0%
K10 modVar int	Ignore	0.004	0.156	0.149	0.528 (0.013)	0.942 (0.006)	100.0%	0.0%
K10 modVar int	Random	0.004	0.156	0.142	0.559 (0.013)	0.928 (0.007)	100.0%	3.1%
K10 noVar bal	Fixed	0.003	0.140	0.142	0.569 (0.013)	0.951 (0.006)	100.0%	0.0%
K10 noVar bal	Ignore	0.003	0.140	0.142	0.568 (0.013)	0.950 (0.006)	100.0%	0.0%
K10 noVar bal	Random	0.003	0.140	0.141	0.573 (0.013)	0.950 (0.006)	100.0%	55.7%
K30 hiVar bal	Fixed	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar bal	Ignore	0.001	0.082	0.099	0.898 (0.008)	0.980 (0.004)	100.0%	0.0%
K30 hiVar bal	Random	0.001	0.082	0.082	0.947 (0.006)	0.937 (0.006)	100.0%	0.0%
K30 hiVar imbal	Fixed	0.004	0.081	0.082	0.974 (0.004)	0.947 (0.006)	100.0%	0.0%
K30 hiVar imbal	Ignore	0.004	0.081	0.098	0.919 (0.007)	0.975 (0.004)	100.0%	0.0%
K30 hiVar imbal	Random	0.004	0.081	0.082	0.974 (0.004)	0.948 (0.006)	100.0%	0.0%
K30 hiVar int	Fixed	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int	Ignore	0.003	0.092	0.100	0.879 (0.008)	0.972 (0.004)	100.0%	0.0%
K30 hiVar int	Random	0.003	0.092	0.082	0.937 (0.006)	0.928 (0.007)	100.0%	0.0%
K30 hiVar int imb	Fixed	0.003	0.100	0.082	0.925 (0.007)	0.894 (0.008)	100.0%	0.0%
K30 hiVar int imb	Ignore	0.003	0.101	0.099	0.855 (0.009)	0.941 (0.006)	100.0%	0.0%
K30 hiVar int imb	Random	0.003	0.100	0.082	0.924 (0.007)	0.895 (0.008)	100.0%	0.0%
K30 modVar bal	Fixed	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.0%
K30 modVar bal	Ignore	0.001	0.079	0.086	0.954 (0.005)	0.963 (0.005)	100.0%	0.0%
K30 modVar bal	Random	0.001	0.079	0.082	0.963 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar imbal	Fixed	0.000	0.081	0.082	0.959 (0.005)	0.956 (0.005)	100.0%	0.0%
K30 modVar imbal	Ignore	0.000	0.081	0.085	0.948 (0.006)	0.961 (0.005)	100.0%	0.0%
K30 modVar imbal	Random	0.000	0.081	0.082	0.956 (0.005)	0.954 (0.005)	100.0%	0.1%
K30 modVar int	Fixed	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 modVar int	Ignore	-0.002	0.091	0.086	0.918 (0.007)	0.936 (0.006)	100.0%	0.0%
K30 modVar int	Random	-0.002	0.091	0.082	0.936 (0.006)	0.923 (0.007)	100.0%	0.0%
K30 noVar bal	Fixed	0.000	0.082	0.082	0.959 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Ignore	0.000	0.082	0.082	0.960 (0.005)	0.943 (0.006)	100.0%	0.0%
K30 noVar bal	Random	0.000	0.082	0.081	0.960 (0.005)	0.943 (0.006)	100.0%	52.7%

3.2 Summary of performance metrics

3.3 Bias

3.4 Power and coverage

3.5 Impact of imbalanced site sizes

3.6 Type I error under the null

Coverage of a 95% CI centered on the true alternative ($\delta = 0.30$, Table 1) is not the same quantity as the Type I error rate of the associated test under a true null effect ($\delta = 0$), which is the quantity most directly relevant to the stratified-

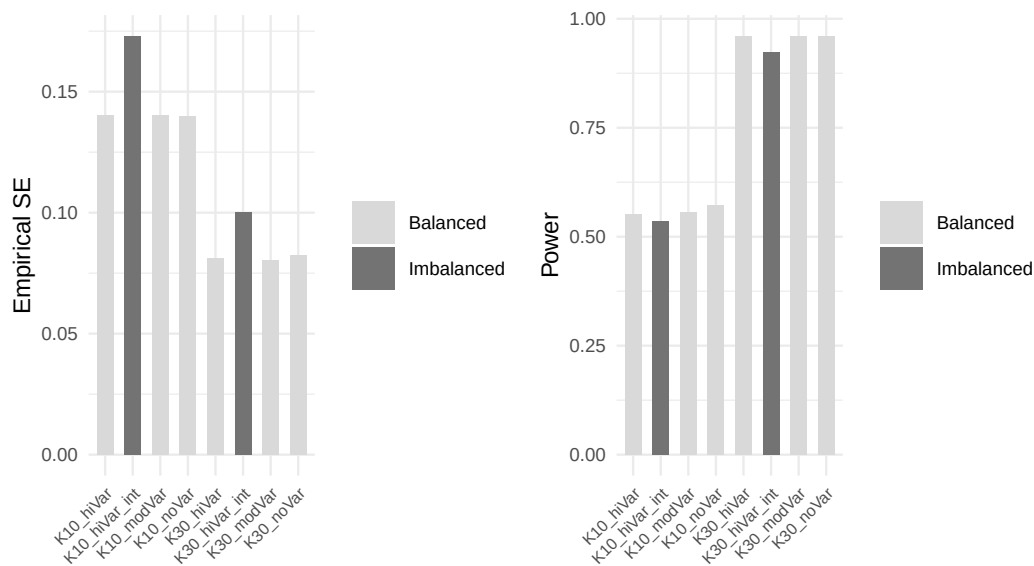


Figure 3: Comparison of balanced versus imbalanced site sizes for the random effects method. Panels show empirical standard error (left) and power (right).

randomization literature motivating this study^{11,13}. Table 2 reports the empirical
rejection rate at $\alpha = 0.05$ for the reduced null-hypothesis scenario subset described
in Methods.

Table 2: Empirical Type I error under the null hypothesis ($\delta = 0$) at nominal $\alpha = 0.05$, by scenario and analytic method, with Morris et al. (2019) Table 6 Monte Carlo SE in parentheses. R = 1500 replications per scenario. ‘Sing.’ is the random-site singular/boundary-fit rate among converged fits.

Scenario	Method	Bias	Type I error (MCSE)	Coverage (MCSE)	Sing.
K10 hiVar bal	Fixed	0.000	0.051 (0.006)	0.949 (0.006)	0.0%
K10 hiVar bal	Ignore	0.000	0.021 (0.004)	0.979 (0.004)	0.0%
K10 hiVar bal	Random	0.000	0.051 (0.006)	0.949 (0.006)	0.1%
K10 hiVar imbal	Fixed	-0.005	0.048 (0.006)	0.952 (0.006)	0.0%
K10 hiVar imbal	Ignore	-0.005	0.019 (0.003)	0.981 (0.003)	0.0%
K10 hiVar imbal	Random	-0.005	0.046 (0.005)	0.954 (0.005)	0.2%
K10 modVar bal	Fixed	0.003	0.057 (0.006)	0.943 (0.006)	0.0%
K10 modVar bal	Ignore	0.003	0.045 (0.005)	0.955 (0.005)	0.0%
K10 modVar bal	Random	0.003	0.057 (0.006)	0.943 (0.006)	3.3%
K10 noVar bal	Fixed	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Ignore	0.001	0.041 (0.005)	0.959 (0.005)	0.0%
K10 noVar bal	Random	0.001	0.041 (0.005)	0.959 (0.005)	56.9%
K30 hiVar bal	Fixed	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar bal	Ignore	0.001	0.014 (0.003)	0.986 (0.003)	0.0%
K30 hiVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	0.0%
K30 hiVar imbal	Fixed	-0.004	0.065 (0.006)	0.935 (0.006)	0.0%
K30 hiVar imbal	Ignore	-0.004	0.023 (0.004)	0.977 (0.004)	0.0%
K30 hiVar imbal	Random	-0.004	0.068 (0.007)	0.932 (0.007)	0.0%
K30 modVar bal	Fixed	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 modVar bal	Ignore	0.000	0.042 (0.005)	0.958 (0.005)	0.0%
K30 modVar bal	Random	0.000	0.052 (0.006)	0.948 (0.006)	0.0%
K30 noVar bal	Fixed	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Ignore	0.001	0.047 (0.005)	0.953 (0.005)	0.0%
K30 noVar bal	Random	0.001	0.048 (0.006)	0.952 (0.006)	53.9%

All three methods maintain Type I error close to the nominal 5% level across the

null-hypothesis scenarios, including the high-site-variance and imbalanced conditions, and including the scenarios in which the random-site model's singular-fit rate is highest. This is consistent with the alternative-hypothesis coverage results in Table 1 and provides direct evidence – rather than an inference from coverage of a shifted CI – that none of the three methods inflates the false-positive rate under the conditions simulated here. It does not, by itself, establish that the t -based reference distribution used for the random-site method (Methods, “Analytic methods”) is correctly calibrated in general; the boundary/singular fits flagged in Table 1 and Table 2 return a point estimate and standard error that are numerically valid but whose sampling distribution at the parameter-space boundary is not guaranteed to match the naive t approximation used here, and this simulation's null-arm results should not be read as validating that approximation outside the specific scenarios simulated.

4 Discussion

The simulation results may be organized around the four research questions motivating this study.

Modeling site effects. Consistent with prior simulation work^{7,8}, we find that the random effects model attained nominal or near- nominal coverage and the highest power across nearly all scenarios. The fixed effects model performed comparably when the number of sites was small ($K = 10$) but lost efficiency with $K = 30$ due to the large number of nuisance parameters. Ignoring site effects produced unbiased point estimates but inflated standard errors when site-level variance was present, leading to conservative inference and reduced power. These findings support the random effects model as a robust default for multicenter RCTs, as recommended in the literature^{1,16}. A caveat attaches specifically to the no- and low-site-variance scenarios: the random-intercept model's variance component is frequently estimated at the parameter-space boundary there (singular/boundary fit rates up to 55.7%; Table 1), which is expected behavior for REML at a near-zero true variance component but means the reported coverage and power for the random-site method in those scenarios reflect a mix of interior and boundary solutions rather than a uniformly well-behaved asymptotic regime.

A methodological caveat applies to how this recommendation is justified, not to the simulation's findings themselves. The data-generating mechanism (Methods, “Data generating model”) draws $\alpha_i \sim N(0, \sigma_\alpha^2)$ afresh for every replicate, i.e. it treats sites as an exchangeable sample from a superpopulation of possible sites. This is a standard simulation convention – a data-generating mechanism must generate *some* concrete site-to-site variability to study estimator behavior under it, and doing so via a random draw is a defensible operational choice regardless of one's view on the next point – but the literature is not unanimous that this superpopulation framing is also the correct justification for fitting a random-effects *analysis* model to a real trial. Falissard and Chavance state explicitly that the random-effects interpretation is only warranted when centres have in fact been

randomly selected from a population of centres, a premise essentially never satisfied in practice, where sites are chosen pragmatically for recruitment capacity and logistics^{3,10}. Kahan and Morris instead justify adjusting for site on design grounds – the analysis should reflect the randomization, which was stratified by site – an argument that requires no claim about how sites were sampled and therefore holds regardless of which camp is right about exchangeability^{8,11}. Edgar, Roberts, and Sharples, re-analyzing the 271-centre CRASH-2 trial, take the same design-based route and recommend random intercepts pragmatically, for power and bias reduction, again without invoking a literal superpopulation assumption¹². The recommendation this simulation supports – prefer random effects when the number of sites is not small – is therefore robust across both justifications; what differs is only which argument a reader should reach for to defend it.

Failure to adjust for stratification. The ‘ignore site’ method effectively represents a failure to adjust for the stratification variable (site) used in randomization. The overcoverage and power loss observed under this method confirm the findings of¹¹: when randomization is stratified by center, the analysis must account for center to obtain valid inference. The magnitude of the power penalty depends on the ICC: with no site-level variance, ignoring site is harmless, but as σ_α^2 increases, the penalty becomes substantial.

Treatment-by-site interaction. Introducing random treatment-by-site interaction ($\sigma_\gamma^2 = 0.04$) increased the variability of treatment effect estimates across replications for all methods. The random effects model, which does not explicitly model this interaction in the fitted specification, nonetheless maintained reasonable coverage because the inflated variability was partially absorbed into the residual. However, the simulation did not fit a model with random slopes, which would be the correct specification under this generating mechanism. Jones and colleagues and Fedorov and Jones have argued that some degree of treatment-by-center interaction is ubiquitous, and that the power to detect it is typically inadequate^{4,15}. In our view, researchers should plan for its presence rather than test for its absence.

Imbalanced site sizes. The comparison of balanced versus imbalanced site sizes reveals that for the random effects model, moderate imbalance ($CV \approx 1.1$) produces only a modest increase in empirical standard error and a correspondingly small decrease in power. This finding is consistent with Ruvuna and with Vierron and Giraudeau, who showed that the efficiency loss from unequal site sizes is generally small unless the imbalance is extreme^{18,19}. The more concerning scenario is when site size is confounded with site characteristics – a possibility not simulated here but discussed elsewhere^{15,17}. Under informative site sizes, where larger sites systematically differ from smaller sites, all methods may produce biased estimates of the population-average treatment effect.

Limitations. We should note that this simulation is restricted to continuous outcomes analyzed with normal-theory models. Binary and time-to-event outcomes raise additional complications, including separation in fixed effects logistic regres-

sion with many small sites^{5,20,23}. The simulation does not incorporate dropout, non-compliance, or informative site sizes. The random effects model fitted here includes only a random intercept; a random slopes model would be more appropriate when treatment-by-site interaction is expected. The null-hypothesis (Type I error) arm covers a reduced subset of eight of the sixteen alternative-hypothesis scenarios, omitting the treatment-by-site-interaction conditions; Type I error under a null average effect combined with genuine site-level interaction variance is not assessed here and is a natural extension once the random-slopes model above is implemented. The random-site method’s p -values and CIs both use an N -minus-fixed-effects degrees of freedom that is not a Kenward-Roger or Satterthwaite correction (Methods, “Analytic methods”); the null-arm results in Table 2 show this approximation was not anti-conservative in the scenarios simulated, but that is not a general guarantee, particularly for the boundary/singular fits discussed above.

5 Future research

Several directions warrant further investigation:

1. **Random slopes models.** Extending the simulation to include a random treatment slope ($\delta + \gamma_i$ with γ_i estimated) would clarify when and how much the random slopes model outperforms the random intercept model under genuine treatment heterogeneity.
2. **Non-normal outcomes.** Binary and count outcomes require generalized linear mixed models or GEE, and the relative performance of methods may differ from the continuous case, particularly with sparse data per site^{20,23}.
3. **Informative site sizes.** If site size is associated with site-level treatment effect (e.g., expert centers enroll more patients and also deliver more effective treatment), standard methods may yield biased population-average estimates. Inverse-probability weighting by site or joint modeling of enrollment and outcome merit investigation¹⁷.
4. **Small-sample corrections.** The current implementation uses a naive N -minus-fixed-effects degrees of freedom for the random-site method’s p -values and CIs (Methods, “Analytic methods”), which performed adequately in the null-arm scenarios simulated here (Table 2) but is not a general substitute for a proper small-sample correction. Kenward-Roger or Satterthwaite denominator degrees of freedom corrections (available in `lmerTest`) should be substituted and evaluated across the same scenario grid, particularly for the $K = 10$, high- site-variance conditions where few sites most threaten the adequacy of any df approximation.
5. **Bayesian approaches.** Hierarchical Bayesian models offer partial pooling of site-specific treatment effects, which may outperform frequentist random effects models when the number of sites is very small⁴.

483 6. **Sample size planning.** Tools that incorporate the design effect from mul-
484 ticenter structure, including anticipated ICC and site size imbalance, into
485 prospective sample size calculations would improve trial planning^{19,24}.

486 6 Data and code availability

487 All data reported here are synthetic, generated by `analysis/scripts/sim_multicenter.R`;
488 no external or human-participant data are used. The simulation and reporting
489 code, including this document's source, are version-controlled in the repository
490 containing this report. The simulation is fully reproducible from the fixed seed
491 and RNG kind documented in Methods ("Simulation procedure"); the exact
492 scenario grid used to produce every table and figure in this report is defined in
493 the `sim-run` code chunk of `report.Rmd`.

494 7 Conflict of interest

495 The author declares no conflict of interest. No external funding supported this
496 work.

497 8 References

498 8.1 Morris et al. (2019) ADEMP Compliance

499 We audited this simulation study against the reporting standards proposed
500 by Morris, White, and Crowther²¹. The original audit is recorded at
501 `docs/morris-audit-2026-04-17.md`; a status update following the 2026-08
502 revision described below is appended to that file.

503 **Verdict:** Partially compliant (updated).

504 **Resolved since the original 2026-04-17 audit:**

- 505 • Monte Carlo SEs are now computed for every performance estimate in
506 `summarize_simulation()` (`mcse_bias`, `mcse_empirical_se`, `mcse_mse`,
507 `mcse_model_se`, `mcse_power`, `mcse_coverage`, `mcse_convergence`,
508 `mcse_singular_rate`) and are displayed alongside the corresponding point
509 estimates for power and coverage in Table 1 and Table 2.
- 510 • `R = 1500` is now justified by an explicit Monte Carlo SE derivation (Meth-
511 ods, "Simulation procedure").
- 512 • Non-convergence is no longer conflated with dropped-and-ignored replica-
513 tions; `n_converged` and `convergence_rate` are reported explicitly, and
514 paired comparisons across methods are preserved because every method's
515 row is retained (as NA on non-convergence) rather than filtered out.
- 516 • `RNGkind("L'Ecuyer-CMRG")` is pinned immediately before the single
517 `set.seed()` call.
- 518 • Per-replicate `.Random.seed` values are captured by `run_simulation()` and
519 attached as the `rng_states` attribute of its return value (not yet persisted

520 to a sidecar file on disk, which remains an open item below).

521 **Gaps identified in the 2026-08 revision, not present in the original audit:**

- 522 • Singular/boundary `lmer` fits were, prior to this revision, silently absorbed
523 into “converged” by the warning-suppression logic in `fit_methods()`; this
524 has been fixed by recording `singular` and `conv_warning` as explicit out-
525 comes, but the fix is new as of this revision and has not yet been audited
526 against Morris §5.1 on a wider scenario set than the one simulated here.
- 527 • Coverage previously used a fixed $z = 1.96$ half-width, inconsistent with
528 the t -based p -values used for power; both now use the same per-method
529 degrees of freedom (Methods, “Analytic methods”), but that shared degrees-
530 of-freedom choice is itself a naive approximation for the random-site method,
531 not a Kenward-Roger or Satterthwaite correction (Future research, item 4).
- 532 • The simulation previously provided no null-hypothesis arm and could not
533 support a Type I error claim; a reduced null-arm scenario subset has been
534 added (Results, “Type I error under the null”), but it does not cover the
535 treatment-by-site-interaction conditions (Discussion, “Limitations”).

536 **Still open:**

- 537 • Per-replicate RNG states are captured in memory but not persisted
538 to `analysis/data/derived_data/rng_states.rds` as originally recom-
539 mended, so a specific failing replication from a completed run cannot yet
540 be re-run from disk without re-executing the whole simulation up to that
541 point.
- 542 • `inst/tinytest/test_basic.R` now exercises `generate_trial_data()` and
543 `summarize_simulation()` (see the package test suite), but does not yet
544 cover `fit_methods()`’s singular-fit detection logic against a constructed
545 non-singular case, which would strengthen confidence in the newly added
546 `singular/conv_warning` columns.

547

548 *Rendered on 2026-08-14 at 11:39 PDT.*

549 *Source: ~/prj/res/07-multicenter-rct/multicenterrct/analysis/report/report.Rmd*

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